Cardiac Deformation Lab

Comprehensive Research Analysis & Scientific Biomechanical Report

Research Prototype — Computational Image Analysis

Advanced continuum kinematics, dense optical-flow deformation fields, longitudinal metric kinetics, and mathematical validation benchmarks.

Report Metadata Attribute	Specification
Study Name	Study Cohort Alpha - Phase 2
Study ID	9b5df003-5640-49ee-9d0f-bca36953d19d
Subject Code	SUBJ-A01
Report Identifier	REP-8BB66D44
Generated Date	October 21, 2026 at 05:06:25 UTC
Software Version	v1.0.4
Report Schema Version	v1.0
Total Analyses Evaluated	2

MANDATORY SCIENTIFIC & NON-CLINICAL DISCLAIMER

This document presents computational image-derived measurements generated by a research prototype. The results are not medical diagnoses, prognoses, treatment recommendations, or clinical findings.
Optical-flow displacement represents apparent image motion and may not equal physical tissue motion. All derived metrics (strains, directions, coherence, TSI) are exploratory mathematical descriptors.

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1. Executive Summary

The study contains **2** computational image analyses evaluated under the research workspace pipeline. Of these, **0** are multi-frame cine temporal sequences, and **2** are differential single-pair analyses. Associated manual reference annotations count: **0**. Spatial physical calibration is active on **0** of 2 analyses, while localized Region-of-Interest (ROI) segmentation is enabled on **0** analyses.

Metric	Value	Metric	Value
Total Image Analyses	2	Calibrated Analyses	0 / 2
Temporal Cine Sequences	0	ROI Segmentation Usage	0 / 2
Manual Reference Annotations	0	Average Computational QC Score	96.0 / 100
Total QC Technical Warnings	0	Longitudinal Tracking Available	Yes

2. Study Information

Study Attribute	Recorded Metadata
Study Name	Study Cohort Alpha - Phase 2
Study Unique ID	9b5df003-5640-49ee-9d0f-bca36953d19d
Subject Pseudonym	SUBJ-A01
Creation Timestamp	October 21, 2026 at 05:06:24 UTC
Last Modified Timestamp	October 21, 2026 at 05:06:24 UTC
Study Description	Longitudinal cardiac motion study
Associated Analysis Sessions	6f4c39f4-9343-4197-8e00-beb645d97ac4, ea2fac46-a031-456f-9317-964c1a90e9e3

3. Computational Methodology

The Cardiac Deformation Lab processes cardiac imagery through an automated, deterministic 12-stage computational continuum mechanics pipeline:
Input → **Preprocessing** (Grayscale conversion, Gaussian smoothing, histogram normalization) → **Motion Estimation** (Farnebäck dense optical flow) → **Displacement Calculation** (Euclidean vector magnitude) → **Spatial Strain Tensor** (Infinitesimal Green-Saint-Venant formulation) → **Principal Strains** (Spectral decomposition of eigenvalues) → **Advanced Kinematic Descriptors** (Motion coherence, strain gradient, anisotropy, deformation energy) → **Temporal Analysis** (Inter-frame cine tracking, TSI) → **ROI & Tracking** (Fixed or Lucas-Kanade updated contours) → **Computational Quality Control** (Signal SNR, gradient smoothness, flow bounds) → **Statistical & Longitudinal Analytics**.

4. Mathematical Framework

Displacement Vector:
$\mathbf{d}(x, y) = [u(x, y), v(x, y)]$■
Displacement magnitude: $D = \sqrt{(u^2 + v^2)}$

Displacement Gradient Tensor:
$\nabla\mathbf{d} = [\ [\partial u/\partial x, \partial u/\partial y], [\partial v/\partial x, \partial v/\partial y]\ ]$

Infinitesimal Strain Tensor:
$\varepsilon = 1/2\ (\nabla\mathbf{d} + (\nabla\mathbf{d})$■$)$
$\varepsilon_xx = \partial u/\partial x$
$\varepsilon_yy = \partial v/\partial y$
$\varepsilon_xy = 1/2\ (\partial u/\partial y + \partial v/\partial x)$

Principal Strains & Directions:
Eigenvalues of ε: λ■, λ■ $= ((\varepsilon_xx + \varepsilon_yy)/2) \pm \sqrt{(((\varepsilon_xx - \varepsilon_yy)/2)^2 + \varepsilon_xy^2)}$

Radial and Circumferential Deformation:
$d_r = \mathbf{d} \cdot \mathbf{e}_r$ (projection onto unit radial vector from centroid)
$d_c = \mathbf{d} \cdot \mathbf{e}_c$ (projection onto unit tangential vector)

Spatial Calibration:
Physical displacement = pixel displacement × pixel spacing (mm/px).
Note: Strain represents dimensionless spatial gradient (mm/mm) and remains invariant to isotropic scaling.

5. Input and Preprocessing

Analysis ID	Input Image Pair	Dimensions	Grayscale	Normalization
6f4c39f4	frame_1.png / frame_2.png	N/A	Complete	Normalized (8-bit)
ea2fac46	frame_1.png / frame_2.png	N/A	Complete	Normalized (8-bit)

6. Motion Analysis

Analysis ID	Mean	Max	Min	Std Dev	Units	Method
6f4c39f4	2.1000	4.5000	0.0000	0.4000	px	Farnebäck Dense Flow

Analysis ID	Mean	Max	Min	Std Dev	Units	Method
ea2fac46	2.4000	4.9000	0.0000	0.5000	px	Farnebäck Dense Flow

Figure 1: Estimated Image Displacement Magnitude Field [Metric: Displacement] (Units: px)

[Visual file not found at outputs/disp.png]

Caption: *Figure: Estimated image displacement magnitude computed via dense optical flow.*

7. Strain Analysis

Analysis ID	Mean Exx	Max Exx	Mean Eyy	Max Eyy	Mean Exy	Mean ε■	Max ε■
6f4c39f4	0.0500	N/A	-0.0400	N/A	0.0100	0.0900	0.1800
ea2fac46	0.0600	N/A	-0.0500	N/A	0.0100	0.1100	0.2000

8. Advanced Kinematic Metrics

Analysis ID	Motion Coherence	Strain Gradient	Anisotropy	Strain Entropy	Deform Energy
6f4c39f4	0.8800	N/A	0.3500	N/A	12.4000
ea2fac46	0.8900	N/A	0.3800	N/A	14.2000

9. Computational Deformation Fingerprint

Dimension Key	Biomechanical Descriptor	Raw Scalar	Normalized Value [0-1]
D_mean	Mean Displacement	2.1000	N/A
D_max	Max Displacement	N/A	N/A
epsilon_mean	Mean Strain Magnitude	N/A	N/A
epsilon_max	Peak Strain Magnitude	N/A	N/A
epsilon_principal	Principal Strain (ε■)	0.0900	N/A
anisotropy	Strain Anisotropy	N/A	N/A
entropy	Strain Entropy	N/A	N/A
strain_gradient	Strain Spatial Gradient	N/A	N/A
coherence	Motion Coherence Index	N/A	N/A
TSI	Temporal Stability Index	N/A	N/A

10. Temporal Cine Analysis

Temporal analysis was not available for this study.

11. Region of Interest (ROI) and Tracking Analysis

Important Scientific Protocol: Full-image metrics and localized ROI metrics are evaluated separately. They are never mixed or aggregated into a single indiscriminate field.

No localized ROI segmentation was applied to the analyses in this study.

12. Spatial Calibration

Calibration Status: Displacement remains in image-space pixels.
Displacement values are converted to physical units only when valid spatial calibration is available.

13. Computational Quality Assessment

Notice: *This quality assessment evaluates computational/image-processing conditions and is not a clinical confidence score.*

Analysis ID	Quality Score	Rating Level	Warnings	Primary Warning Message
6f4c39f4	96.0 / 100	High	0	N/A
ea2fac46	N/A / 100	N/A	0	N/A

14. Longitudinal Multi-Session Analysis

Metric Name	Trend Direction	Slope	R^2	Abs Change	Pct Change
Mean Displacement	stable	N/A	N/A	N/A	N/A
Maximum Displacement	stable	N/A	N/A	N/A	N/A
Mean Strain	stable	N/A	N/A	N/A	N/A
Maximum Strain	stable	N/A	N/A	N/A	N/A
Mean Principal Strain	stable	N/A	N/A	N/A	N/A
Maximum Principal Strain	stable	N/A	N/A	N/A	N/A
Anisotropy Index	stable	N/A	N/A	N/A	N/A
Strain Entropy	stable	N/A	N/A	N/A	N/A
Strain Gradient	stable	N/A	N/A	N/A	N/A
Motion Coherence	stable	N/A	N/A	N/A	N/A
Temporal Stability Index	stable	N/A	N/A	N/A	N/A
ROI Mean Displacement	stable	N/A	N/A	N/A	N/A
ROI Mean Strain	stable	N/A	N/A	N/A	N/A
ROI Principal Strain	stable	N/A	N/A	N/A	N/A
Computational Quality Score	stable	N/A	N/A	N/A	N/A

15. Advanced Statistical Analytics

Metric	N	Mean	Median	Std Dev	Min	Max	IQR
Mean Displacement	2	2.2500	2.2500	N/A	2.1000	2.4000	0.1500
Max Displacement	2	4.7000	4.7000	N/A	4.5000	4.9000	0.2000
Mean Strain	2	0.0550	0.0550	N/A	0.0500	0.0600	0.0050
Max Strain	0	N/A	N/A	N/A	N/A	N/A	N/A
Mean Principal Strain	2	0.1000	0.1000	N/A	0.0900	0.1100	0.0100
Max Principal Strain	2	0.1900	0.1900	N/A	0.1800	0.2000	0.0100
Anisotropy Index	2	0.3650	0.3650	N/A	0.3500	0.3800	0.0150
Strain Entropy	0	N/A	N/A	N/A	N/A	N/A	N/A
Strain Gradient	0	N/A	N/A	N/A	N/A	N/A	N/A
Motion Coherence	2	0.8850	0.8850	N/A	0.8800	0.8900	0.0050
Temporal Stability Index	0	N/A	N/A	N/A	N/A	N/A	N/A
Quality Score	1	96.0000	96.0000	N/A	96.0000	96.0000	0.0000

Notice: Correlation describes statistical association and does not imply causation.

16. Synthetic / Phantom Validation Benchmark

Methodological Context: *Synthetic validation evaluates algorithmic behavior against mathematically controlled reference fields and does not establish clinical validity.*

Parameter	Setting	Validation Error Metric	Observed Value
Deformation Mode	N/A	Displacement MAE	N/A px
Deformation Strength	N/A	Displacement RMSE	N/A px
Noise Amplitude	N/A	Displacement Pearson r	N/A
Gaussian Blur Sigma	N/A	Principal Strain MAE	N/A

17. Manual Reference Benchmark

Important Notice: *Manual annotations are reference measurements for computational benchmarking and are subject to annotator variability. They are not clinical ground truth.*

No manual landmark or contour annotations available for reference benchmarking in this study.

18. In-Study Comparison Analysis

Differential comparison could not be calculated.

19. Limitations

1. Apparent Image Motion vs. Physical Motion: Optical flow measures brightness constancy variations in image pixel coordinates, which may differ from physical tissue motion under out-of-plane speckle decorrelation or non-rigid myocardial through-plane deformation.
2. Two-Dimensional Restriction: All strain tensors (ε_xx, ε_yy, ε_xy) are computed in a 2D image plane, omitting radial-longitudinal shear components present in 3D cardiac mechanics.
3. Temporal Sampling Rate: Frame rates affect peak displacement and strain derivative estimation. Nyquist limits govern kinematic precision.
4. Spatial Calibration Constraints: Physical conversions (mm/px) assume isotropic, uniform pixel spacing in orthogonal directions.
5. Manual Annotation Variability: Reference annotations represent subjective operator placements with intra- and inter-observer variance.
6. Statistical Sample Size: Small study cohorts limit statistical generalizability; correlations represent association rather than causation.

20. Computational Conclusion

The analysis demonstrates the extraction of image-derived motion, displacement, strain, and related computational descriptors using the implemented research pipeline. The pipeline successfully processed all associated input frames, computed dense continuum fields, evaluated spatial and temporal consistency, and compiled quantitative multi-session trends.

Scientific Interpretation Notice: The results should be interpreted as computational research outputs and not as clinical findings. These measurements provide an objective quantitative basis for engineering research, computational algorithm validation, and academic inquiry.

21. Appendix & System Metadata

System Component	Technical Specification
Software System	Cardiac Deformation Lab (CDL)
Software Version	v1.0.4
Report Engine	ReportLab PDF Engine with Dynamic Two-Pass Canvas
Report Identifier	REP-8BB66D44
Timestamp of Generation	October 21, 2026 at 05:06:25 UTC
Optical Flow Algorithm	Farnebäck Dense Optical Flow (OpenCV)
Strain Formulation	Infinitesimal Green-Saint-Venant Deformation Tensor
Included Analysis Sessions	6f4c39f4, ea2fac46